

git log/history

References

- [Visualizing Code: Polyglot Notebooks Repository \(YouTube\)](#)
- [gitstractor \(GitHub\)](#)
- [Neo4j Python Driver](#)

Command line execution (CLI mode): Yes

Git History - Directory Commit Statistics

Treemap Layout Functions and Constants

Visualization Data Preparation Functions

File Data Preparation Functions

File Data Preparation

```

Received notification from DBMS server: {severity: WARNING} {code: Neo.ClientNotification.Statement.AggregationSkippedNull} {category: UNRECOGNIZED} {title: The query contains an aggregation function that skips null values.} {description: null value eliminated in set function.} {position: None} for query: "// List git files with commit statistics\n\n MATCH (git_file:File&Git&!Repository)\n WHERE git_file.deletedAt IS NULL // filter out deleted files\n WITH percentileDisc(git_file.createdAtEpoch, 0.5) AS medianCreatedAtEpoch\n ,percentileDisc(git_file.lastModificationAtEpoch, 0.5) AS medianLastModificationAtEpoch\n ,collect(git_file) AS git_files\n UNWIND git_files AS git_file\n WITH *\n ,datetime.fromepochMillis(coalesce(git_file.createdAtEpoch, medianCreatedAtEpoch)) AS fileCreatedAtTimestamp\n ,datetime.fromepochMillis(coalesce(git_file.lastModificationAtEpoch, git_file.createdAtEpoch, medianLastModificationAtEpoch)) AS fileLastModificationAtTimestamp\n MATCH (git_repository:Git&Repository)-[:HAS_FILE]->(git_file)\n MATCH (git_commit:Git&Commit)-[:CONTAINS_CHANGE]->(git_change:Git&Change)-->(old_files_included:Git&File&!Repository)-[:HAS_NEW_NAME*0..3]->(git_file)\n RETURN git_repository.name + '/' + git_file.relativePath AS filePath\n ,split(git_commit.author, ' <')[0] AS author\n ,count(DISTINCT git_commit.sha) AS commitCount\n ,date(max(git_commit.date)) AS lastCommitDate\n ,max(date(fileCreatedAtTimestamp)) AS lastCreationDate\n ,max(date(fileLastModificationAtTimestamp)) AS lastModificationDate\n ,duration.inDays(date(max(git_commit.date)), date()).days AS daysSinceLastCommit\n ,duration.inDays(max(fileCreatedAtTimestamp), datetime()).days AS daysSinceLastCreation\n ,duration.inDays(max(fileLastModificationAtTimestamp), datetime()).days AS daysSinceLastModification\n AS maxCommitSha\n ORDER BY filePath ASCENDING, commitCount DESCENDING"

```

Data Preview

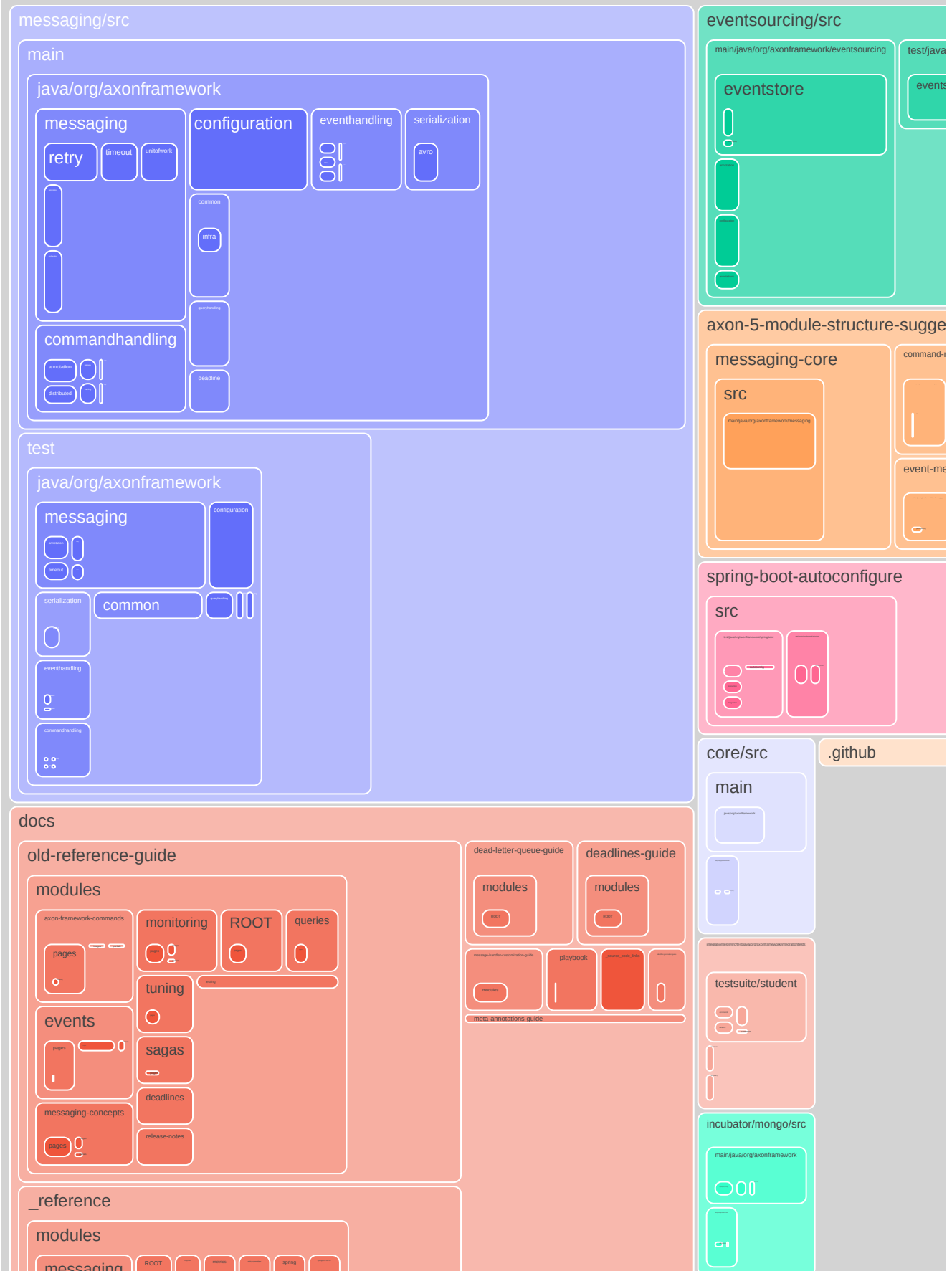
	fileCount	authorCount	commitCount	daysSinceLastCommit	daysSinceLastCreation	daysSinceLastModification
count	305.000000	305.000000	305.000000	305.000000	305.000000	305.000000
mean	14.062295	7.636066	348.750820	412.901639	651.872131	583.134426
std	57.455641	7.093502	1465.259193	1048.994357	1222.558738	1231.807657
min	1.000000	2.000000	2.000000	0.000000	0.000000	0.000000
25%	1.000000	3.000000	24.000000	18.000000	55.000000	17.000000
50%	4.000000	6.000000	75.000000	42.000000	262.000000	104.000000
75%	8.000000	8.000000	187.000000	250.000000	347.000000	292.000000
max	904.000000	71.000000	22225.000000	4791.000000	4879.000000	4790.000000

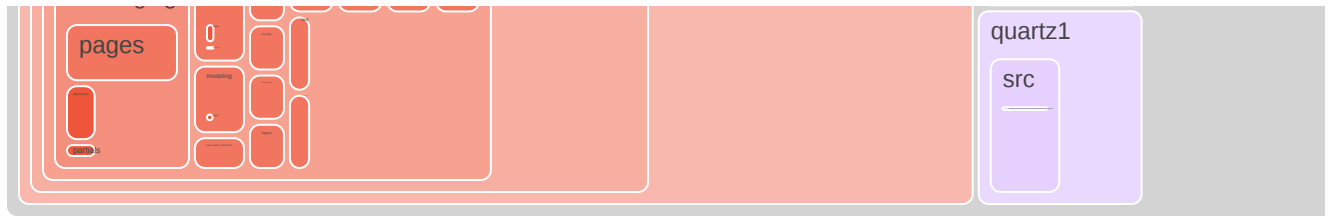
	directoryPath	directoryParentPath
0	AxonFramework-4.11.1/.idea/copyright	AxonFramework-4.11.1/.idea
1	AxonFramework-4.11.1/.idea/inspectionProfiles	AxonFramework-4.11.1/.idea
2	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src/test/java/org/axonframework/messaging	AxonFramework-4.11.1/axon-5-module-structure-suggestion/messaging-core/src test/java/org/
3	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector/command	AxonFramework-4.11.1/axon-server-connector/src/main/java/org/axonframework/axonserver/connector
4	AxonFramework-4.11.1/core/src/test/java/org/axonframework/commandhandling/disruptor	AxonFramework-4.11.1/core/src/test/java/org/axonframework
5	AxonFramework-4.11.1/core/src/test/java/org/axonframework/contextsupport/spring	AxonFramework-4.11.1/core/src/test/java/org/axonframework
6	AxonFramework-4.11.1/core/src/test/java/org/axonframework/eventhandling/scheduling/quartz	AxonFramework-4.11.1/core/src/test/java/org/axonframework eventl
7	AxonFramework-4.11.1/docs/_playbook/localLinks	AxonFramework-4.11.1/docs/_playbook
8	AxonFramework-4.11.1/docs/_reference/modules/ROOT/attachments	AxonFramework-4.11.1/docs/_reference/modules/ROOT
9	AxonFramework-4.11.1/docs/_reference/modules/ROOT/partial	AxonFramework-4.11.1/docs/_reference/modules/ROOT
10	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/pages	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
11	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector/partial	AxonFramework-4.11.1/docs/_reference/modules/axon-server-connector
12	AxonFramework-4.11.1/docs/_reference/modules/configuration/pages	AxonFramework-4.11.1/docs/_reference/modules/configuration
13	AxonFramework-4.11.1/docs/_reference/modules/configuration/partial	AxonFramework-4.11.1/docs/_reference/modules/configuration
14	AxonFramework-4.11.1/docs/_reference/modules/disruptor/pages	AxonFramework-4.11.1/docs/_reference/modules/disruptor
15	AxonFramework-4.11.1/docs/_reference/modules/disruptor/partial	AxonFramework-4.11.1/docs/_reference/modules/disruptor
16	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/pages	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
17	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing/partial	AxonFramework-4.11.1/docs/_reference/modules/eventsourcing
18	AxonFramework-4.11.1/docs/_reference/modules/legacy/pages	AxonFramework-4.11.1/docs/_reference/modules/legacy
19	AxonFramework-4.11.1/docs/_reference/modules/legacy/partial	AxonFramework-4.11.1/docs/_reference/modules/legacy
20	AxonFramework-4.11.1/docs/_reference/modules/metrics/pages	AxonFramework-4.11.1/docs/_reference/modules/metrics
21	AxonFramework-4.11.1/docs/_reference/modules/metrics/partial	AxonFramework-4.11.1/docs/_reference/modules/metrics
22	AxonFramework-4.11.1/docs/_reference/modules/micrometer/pages	AxonFramework-4.11.1/docs/_reference/modules/micrometer
23	AxonFramework-4.11.1/docs/_reference/modules/micrometer/partial	AxonFramework-4.11.1/docs/_reference/modules/micrometer
24	AxonFramework-4.11.1/docs/_reference/modules/modeling/partial	AxonFramework-4.11.1/docs/_reference/modules/modeling
25	AxonFramework-4.11.1/docs/_reference/modules/spring/pages	AxonFramework-4.11.1/docs/_reference/modules/spring
26	AxonFramework-4.11.1/docs/_reference/modules/spring/partial	AxonFramework-4.11.1/docs/_reference/modules/spring
27	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/pages	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter
28	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter/partial	AxonFramework-4.11.1/docs/_reference/modules/springboot-starter
29	AxonFramework-4.11.1/docs/_reference/modules/test/pages	AxonFramework-4.11.1/docs/_reference/modules/test

Number of files per directory

Directories and their file count

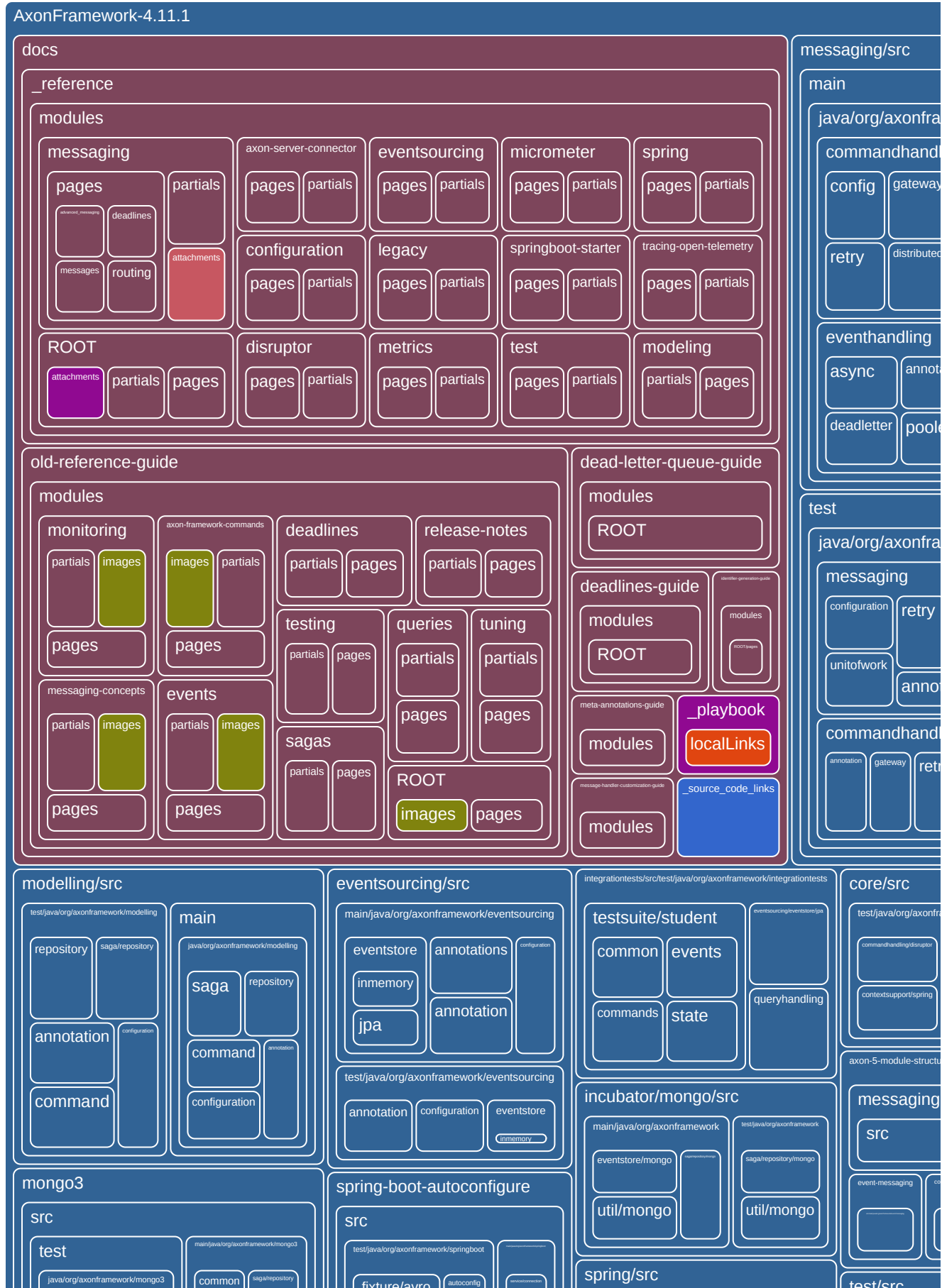
AxonFramework-4.11.1

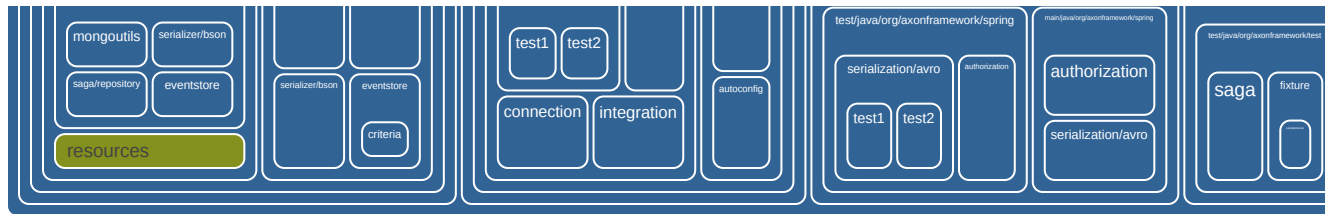




Most frequent file extension per directory

Most frequent file extension per directory

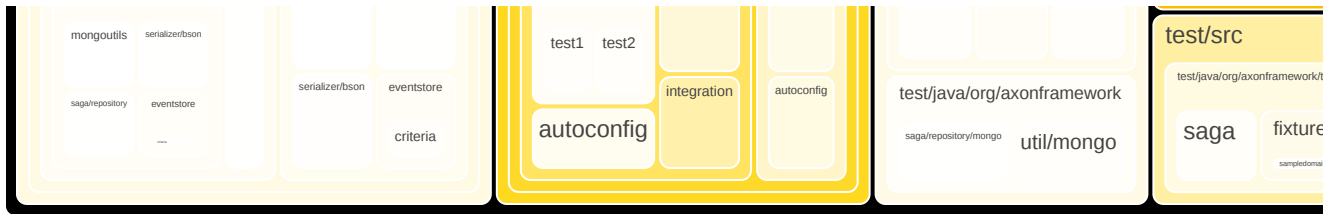




Number of commits per directory

Number of git commits

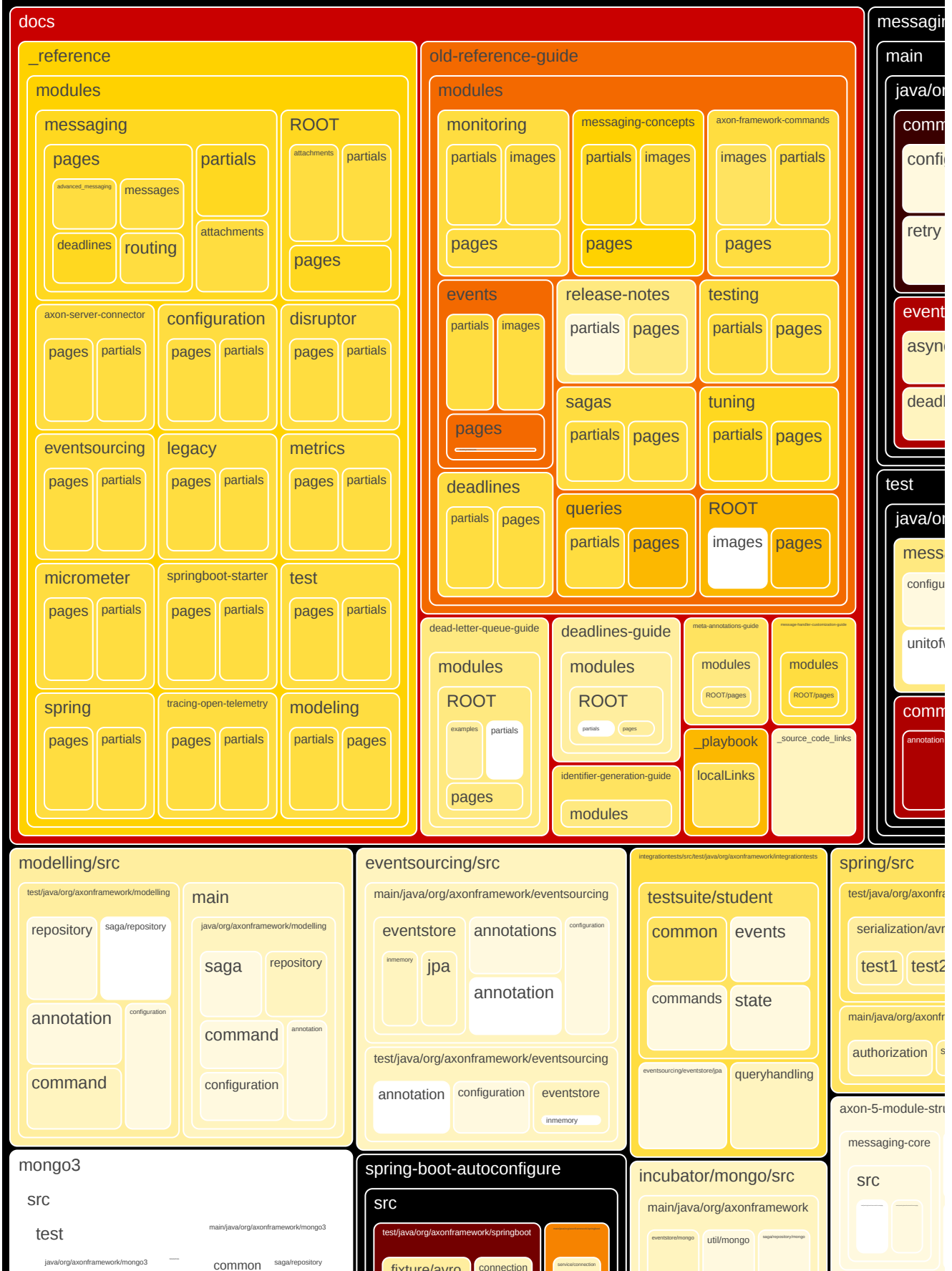


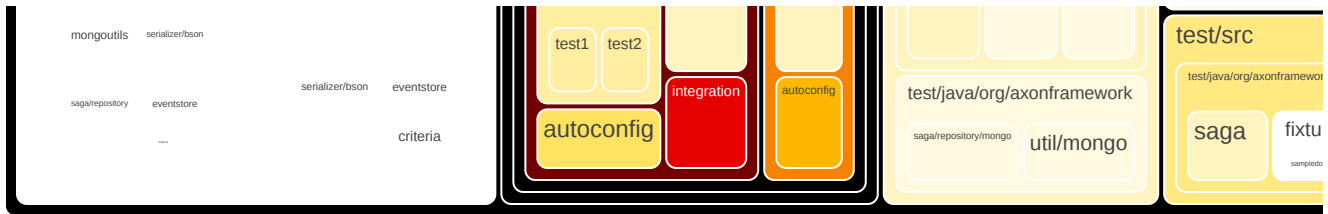


Number of distinct authors per directory

Number of distinct commit authors

AxonFramework-4.11.1

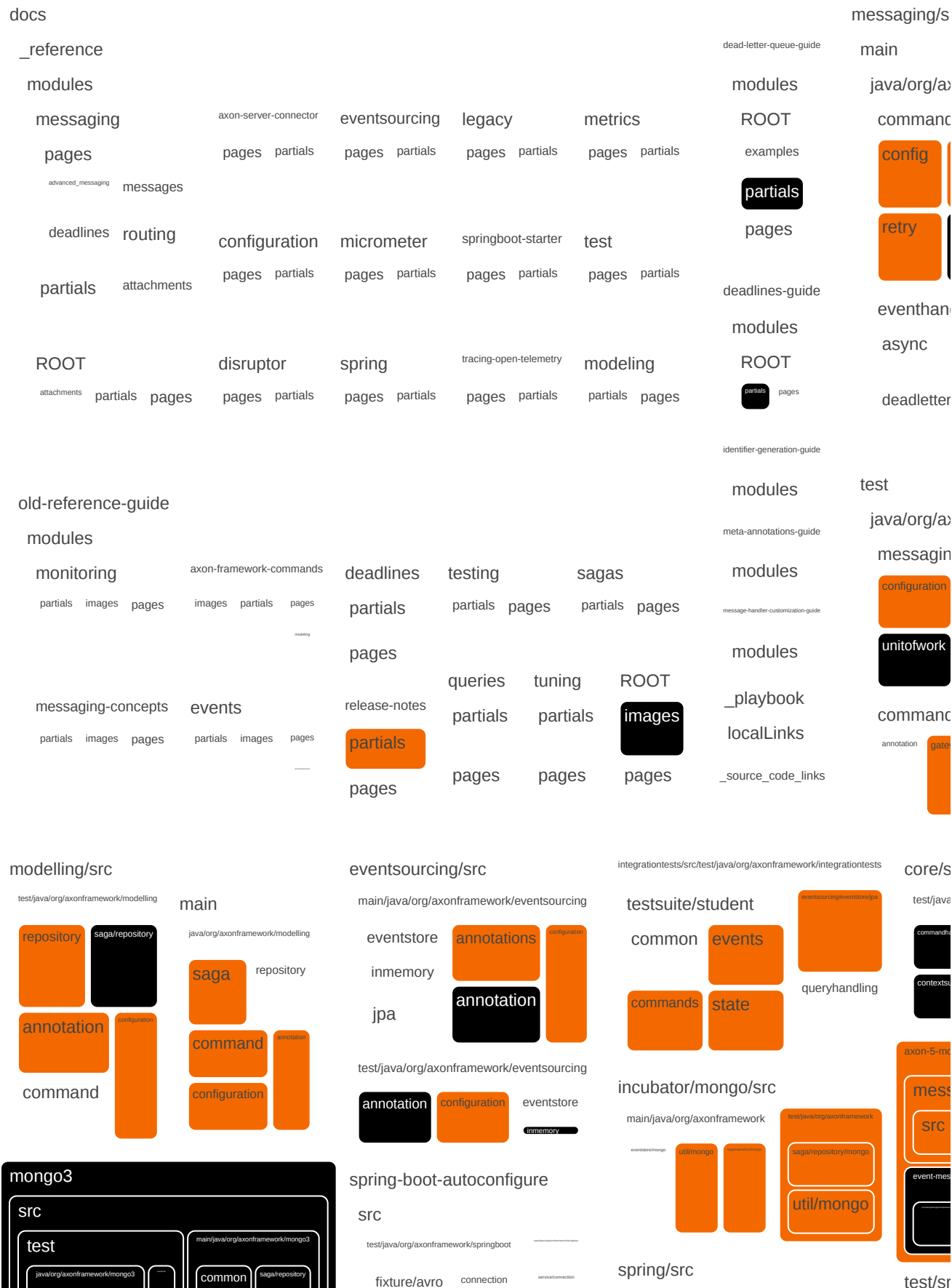


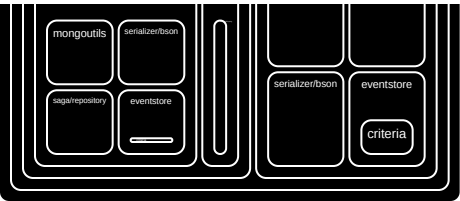


Directories with very few different authors

Number of distinct commit authors (red/black = only one or very few authors)

AxonFramework-4.11.1

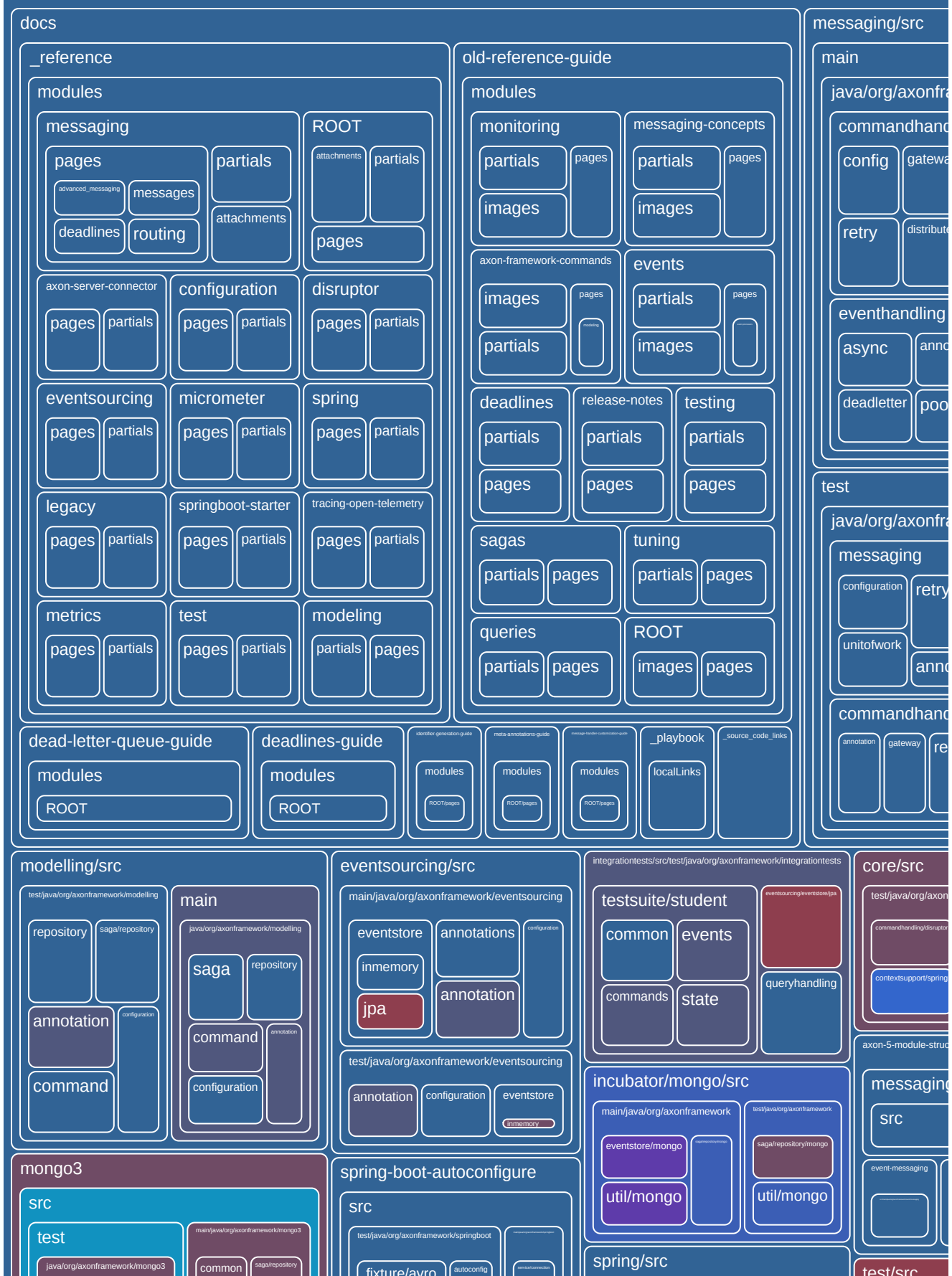


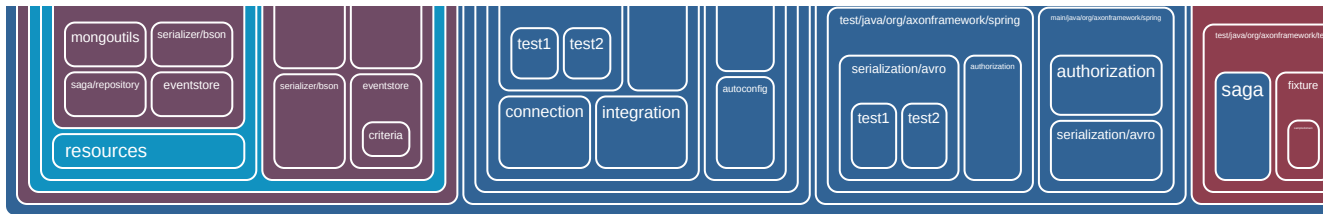


Main author per directory

Main authors with highest number of commits

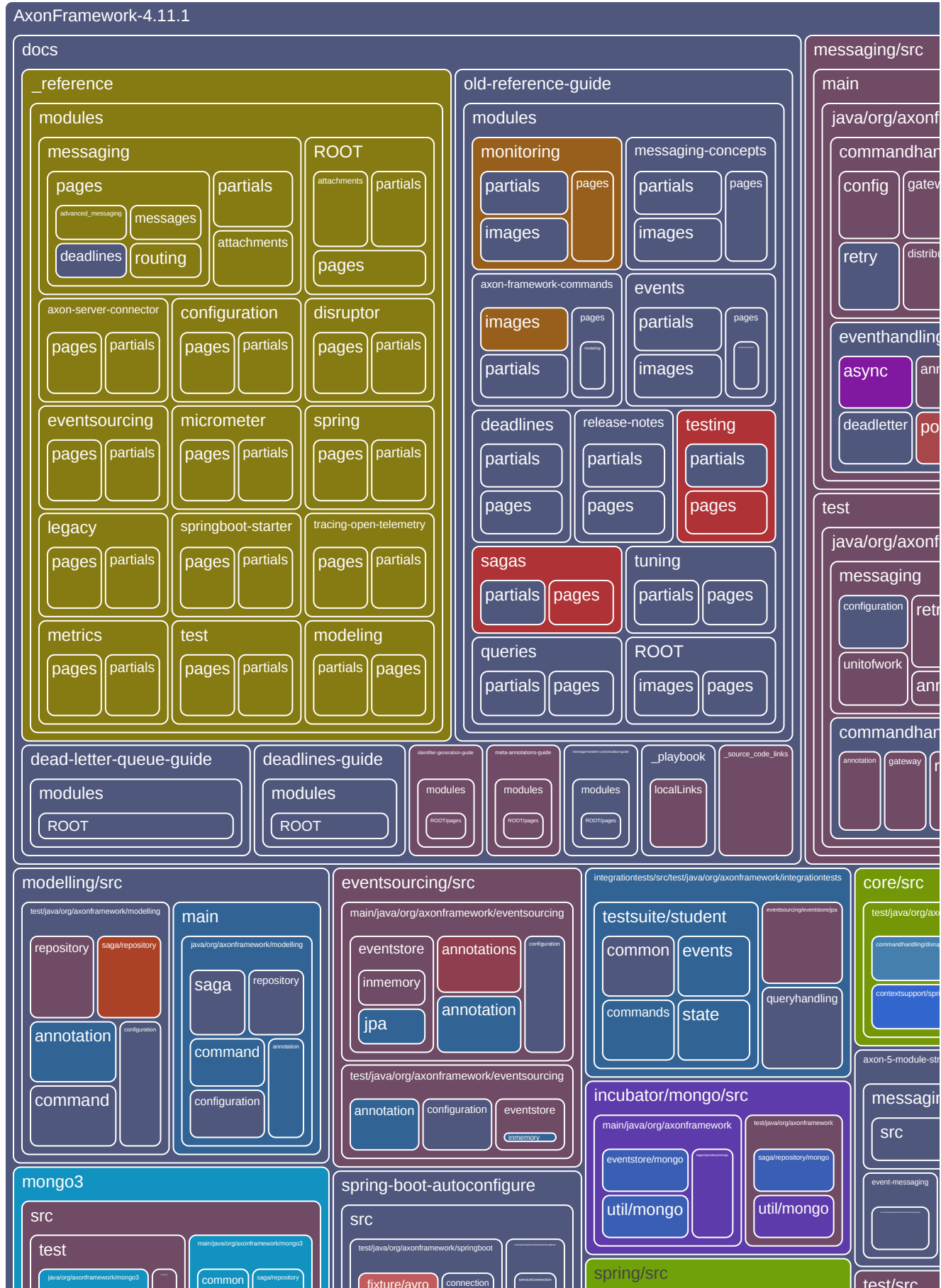
AxonFramework-4.11.1

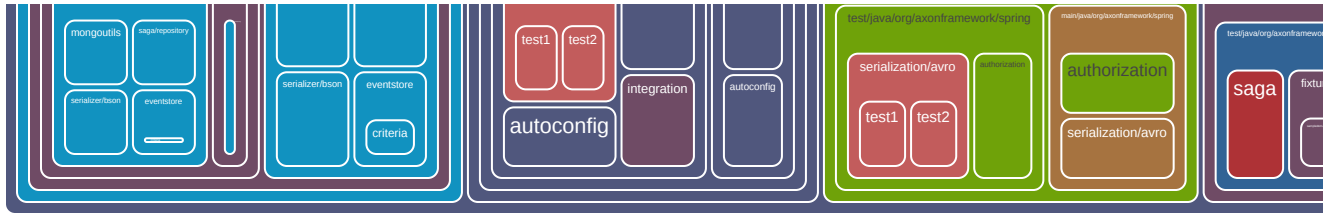




Second author per directory

Second author with the second highest number of commits



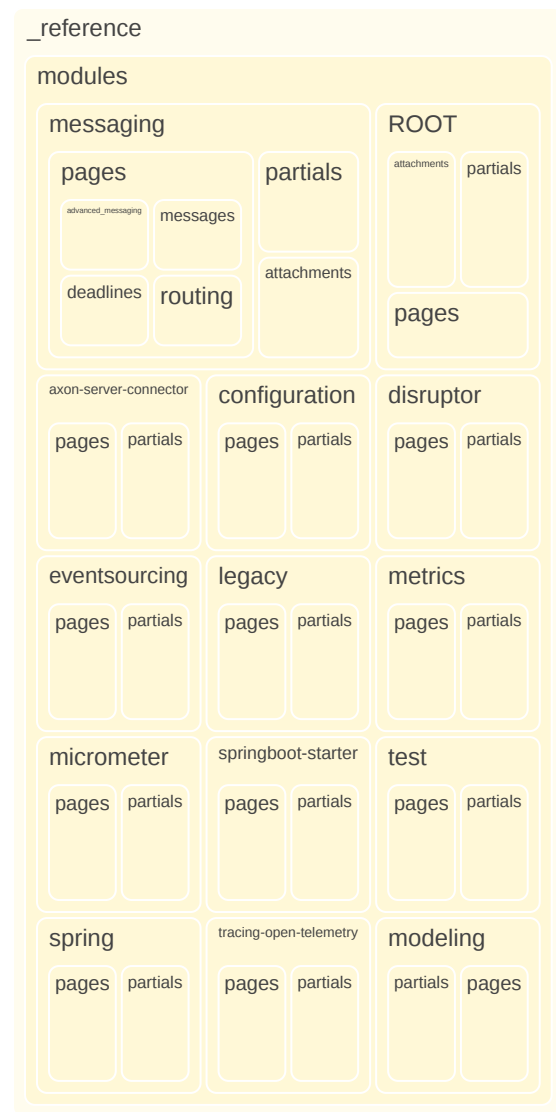


Days since last commit per directory

Days since last commit

AxonFramework-4.11.1

docs



old-reference-guide



messaging

main

java/org

comm

config

retry

eventh

async

deadle

test

java/org

messa

configura

unitofw

comm

annotation

modelling/src



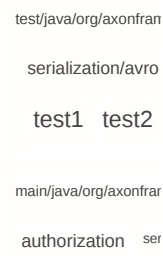
eventsourcing/src



integrationtests/src/test/java/org/axonframework/integrationtests



spring/src



axon-5-module-struct

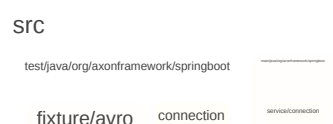
messaging-core

src

mongo3



spring-boot-autoconfigure



incubator/mongo/src



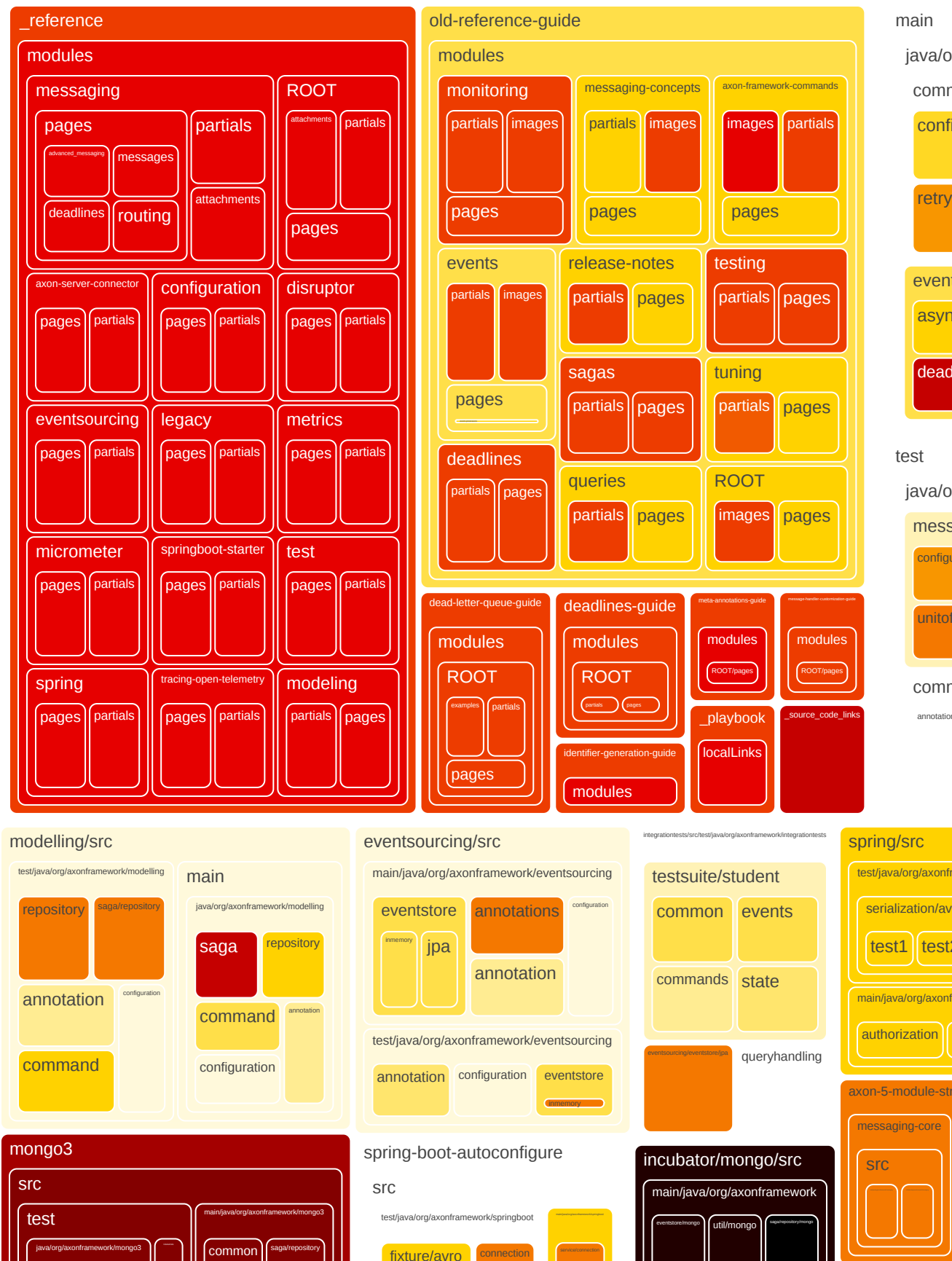


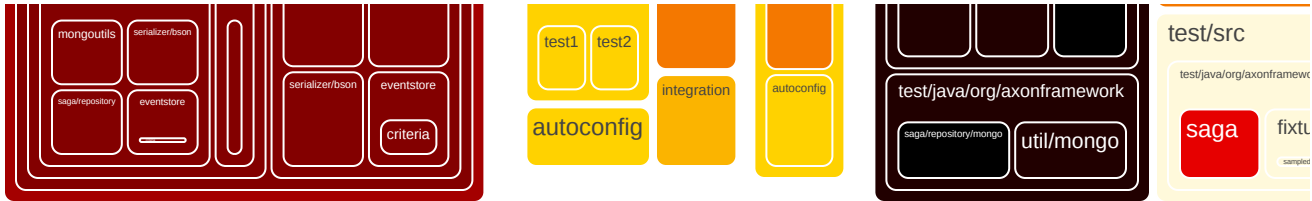
Days since last commit per directory (ranked)

Rank of days since last commit

AxonFramework-4.11.1

docs



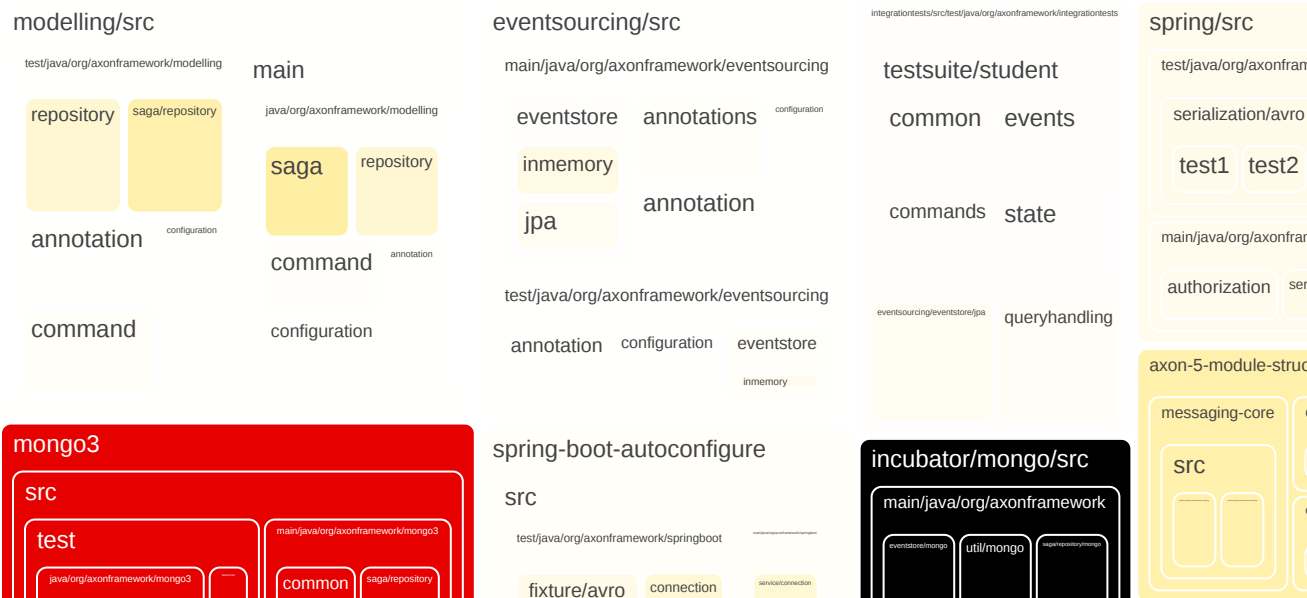
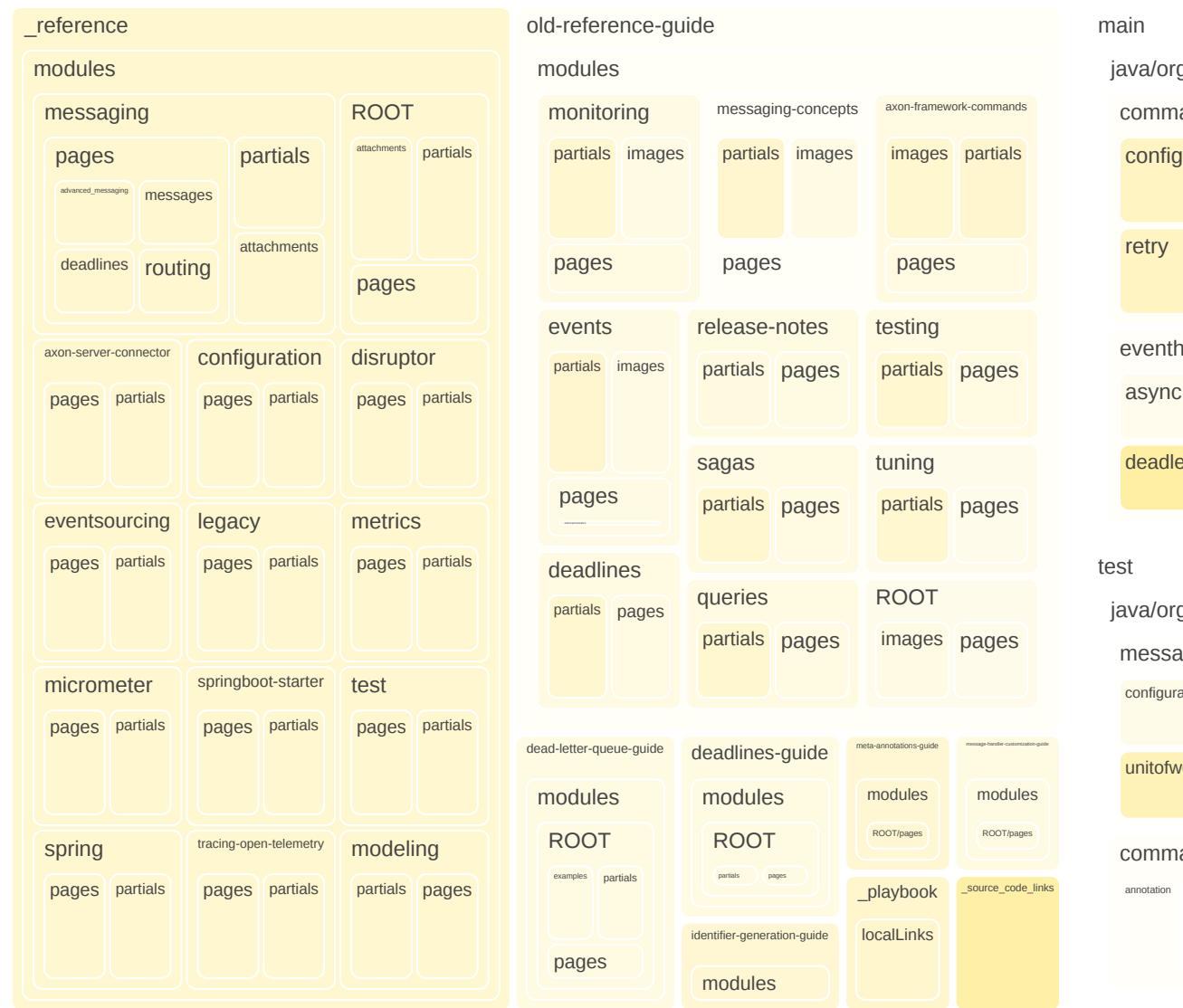


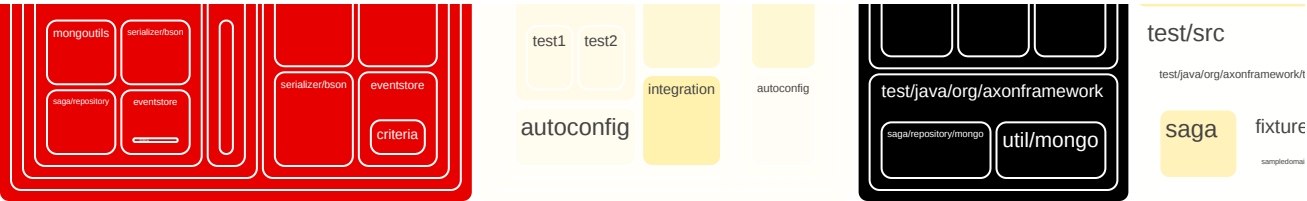
Days since last file creation per directory

Days since last file creation

AxonFramework-4.11.1

docs



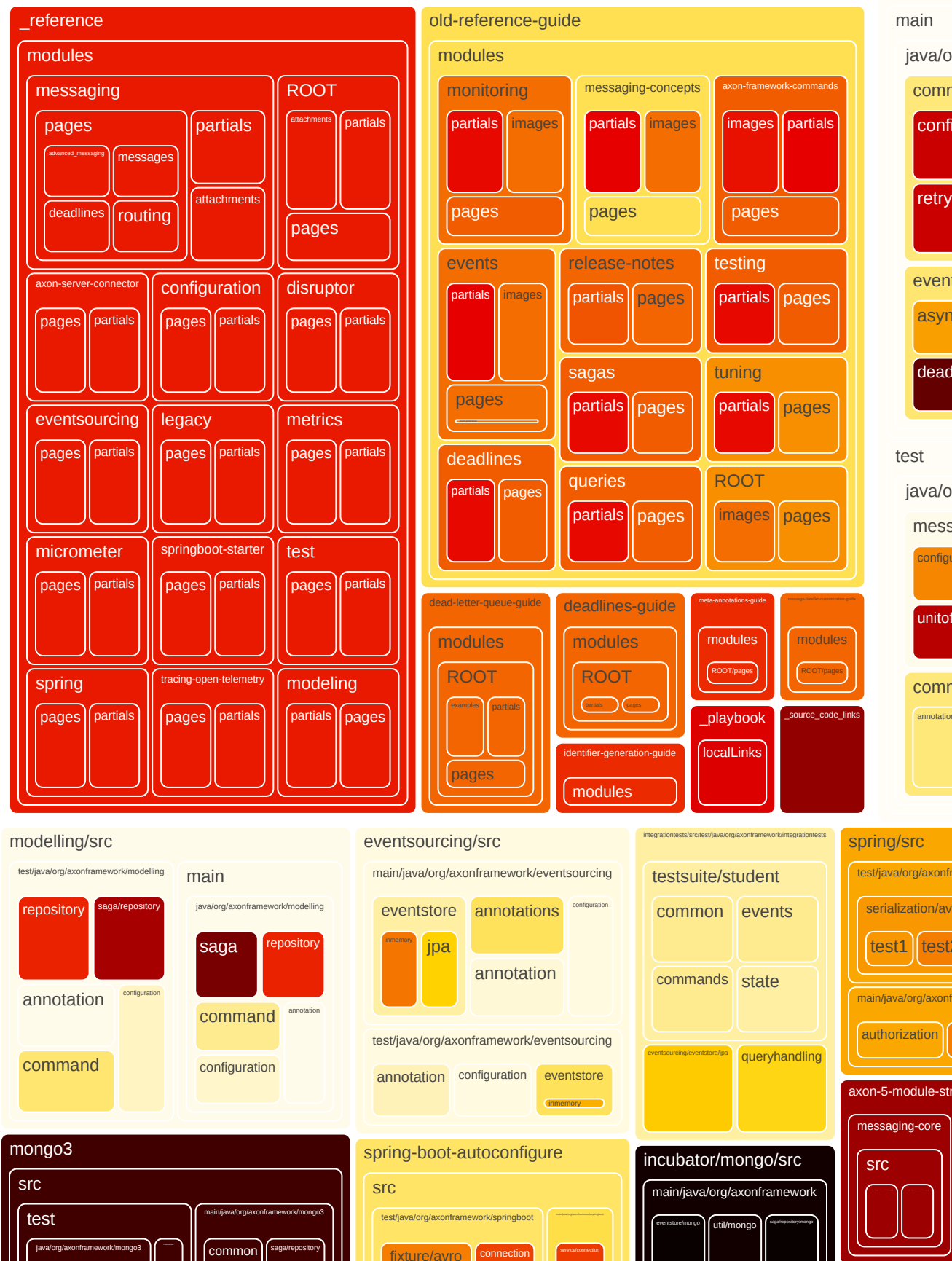


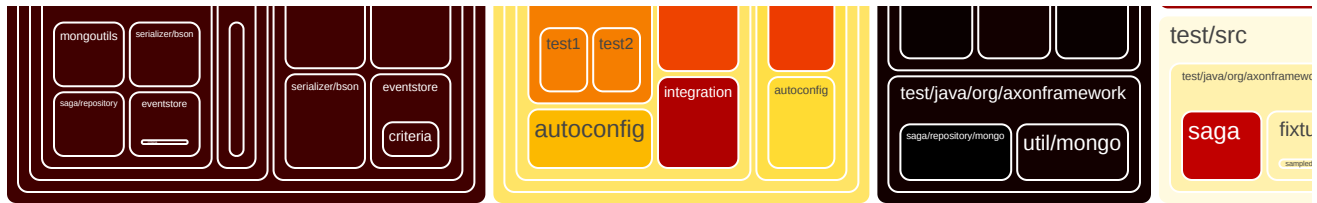
Days since last file creation per directory (ranked)

Rank of days since last file creation

AxonFramework-4.11.1

docs



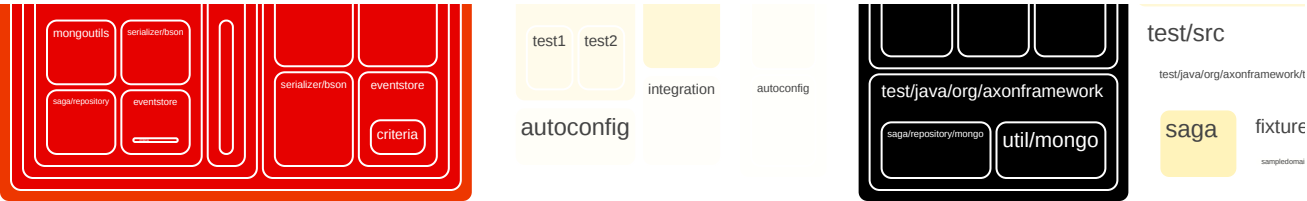


Days since last file modification per directory

Days since last file modification

AxonFramework-4.11.1



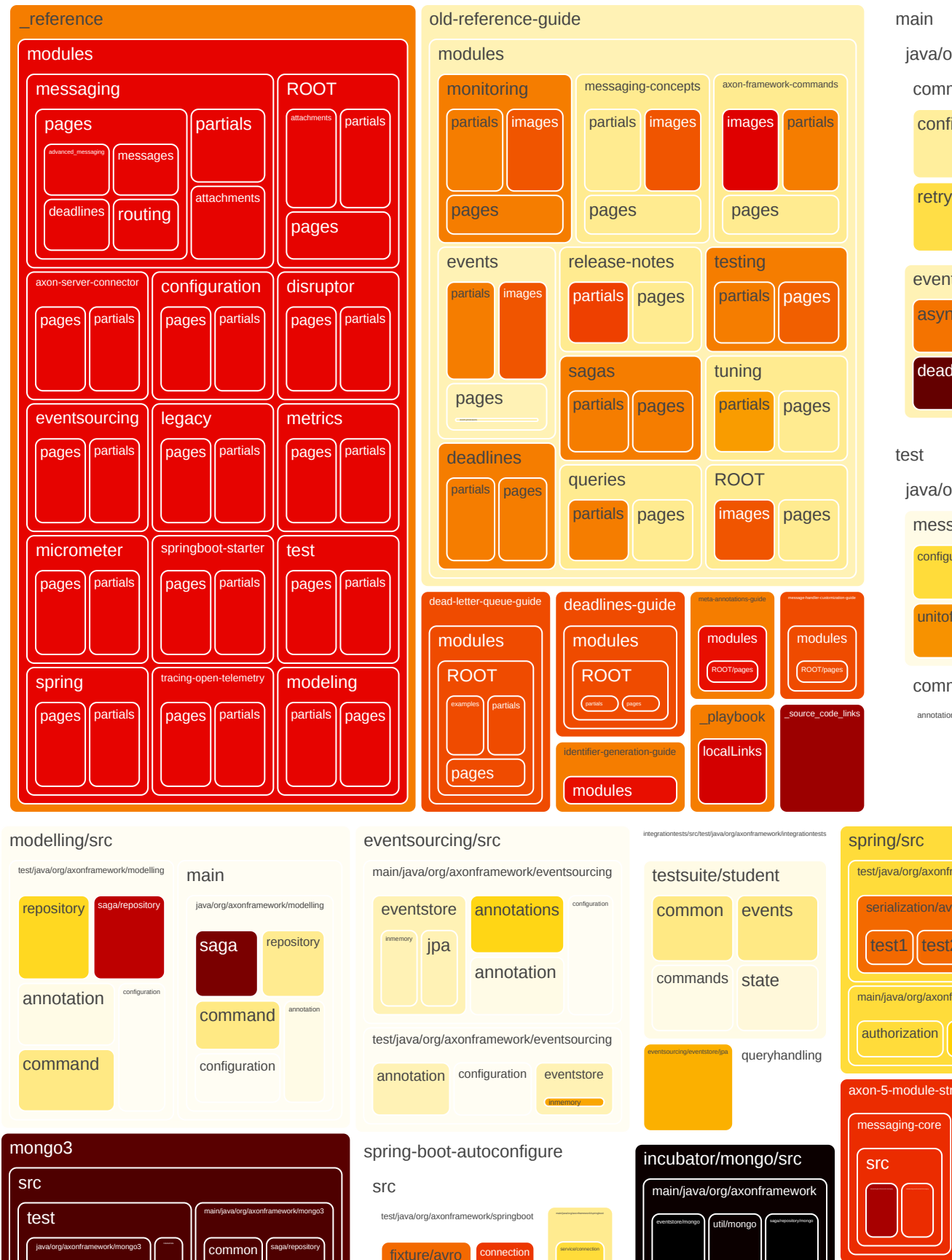


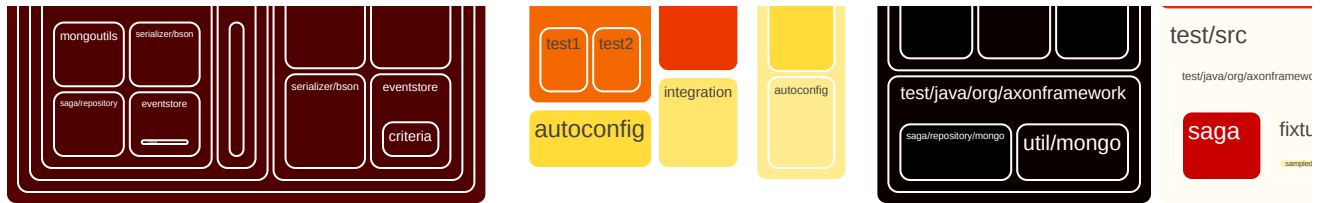
Days since last file modification per directory (ranked)

Rank of days since last file modification

AxonFramework-4.11.1

docs

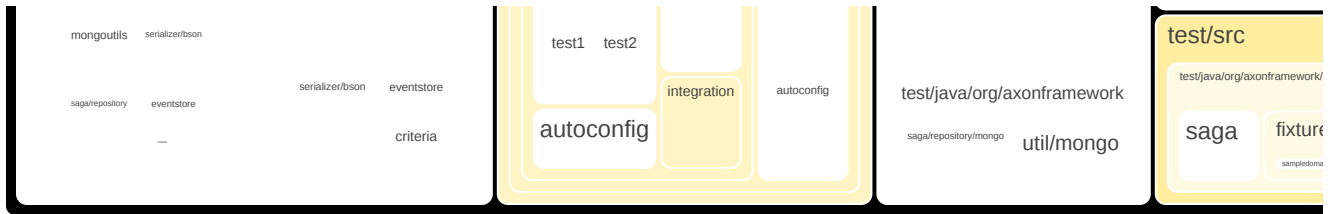




File changed frequently with other files

Pairwise file changes





Filecount per commit

Shows how many commits had changed one file, how many had changed two files, and so on. The chart is limited to 30 lines for improved readability. The data preview also includes overall statistics including the number of commits that are filtered out in the chart.

Preview data

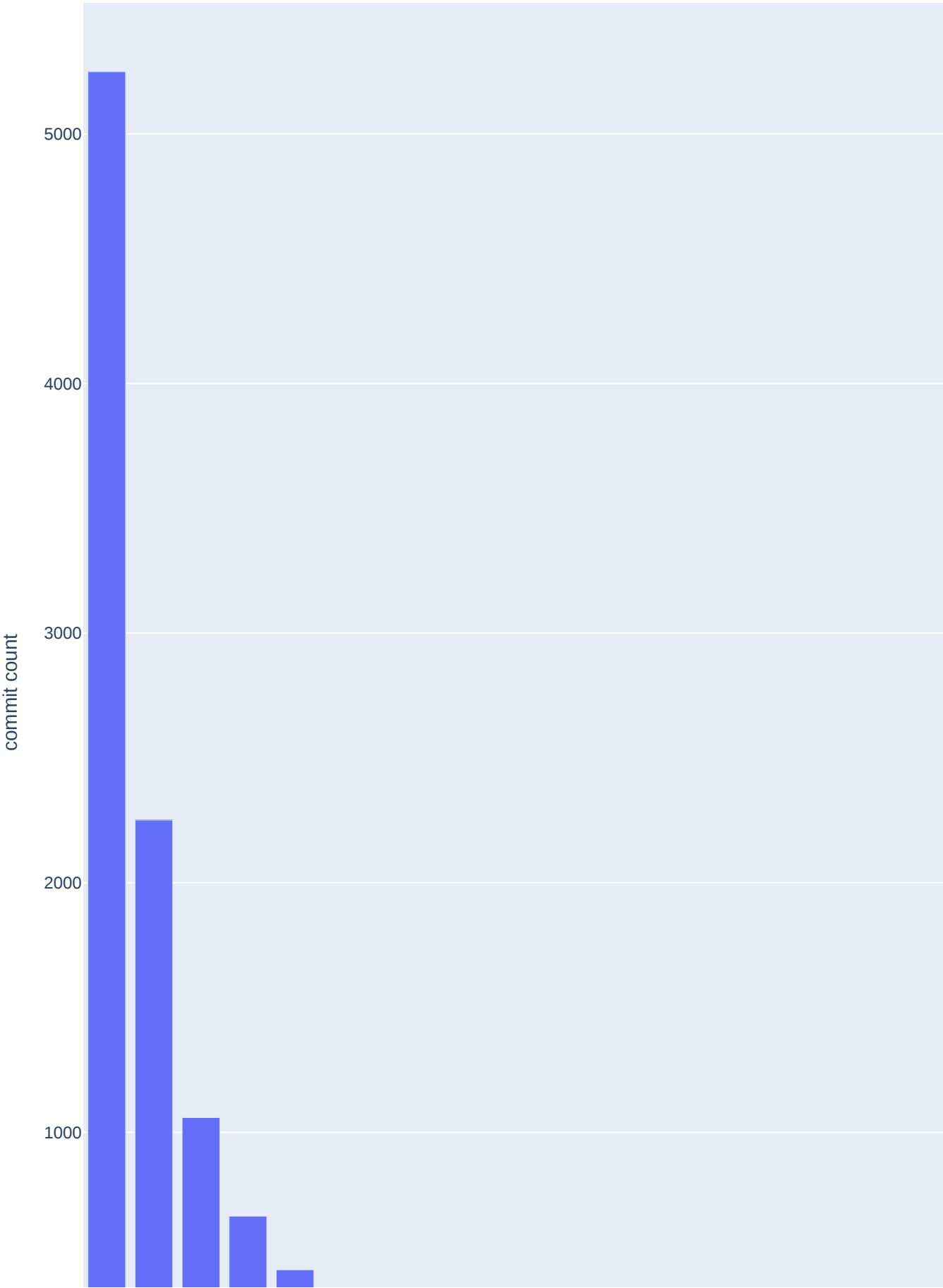
Sum of commits that changed more than 30 files (each) = 1181
Max changed files with one commit = 4041

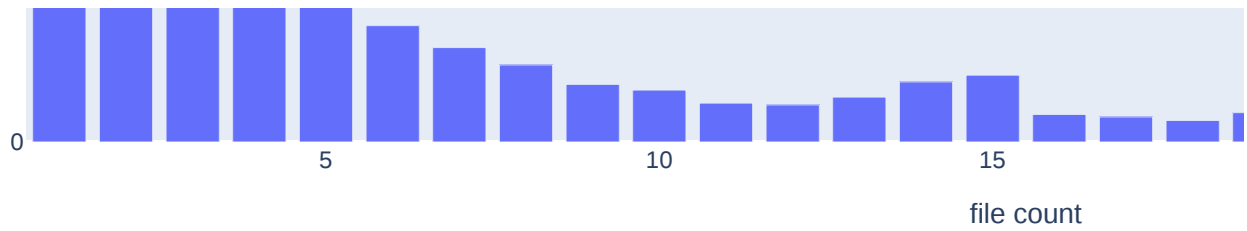
	filesPerCommit	commitCount
count	377.000000	377.000000
mean	398.572944	35.607427
std	559.546155	302.200493
min	1.000000	1.000000
25%	95.000000	1.000000
50%	204.000000	2.000000
75%	429.000000	5.000000
max	4041.000000	5249.000000

	filesPerCommit	commitCount
0	1	5249
1	2	2251
2	3	1059
3	4	664
4	5	449
5	6	329
6	7	267
7	8	218
8	9	163
9	10	147
10	11	110
11	12	105
12	13	127
13	14	170
14	15	189
15	16	78
16	17	71
17	18	61
18	19	83
19	20	74
20	21	46
21	22	68
22	23	34
23	24	38
24	25	43
25	26	33
26	27	34
27	28	30
28	29	29
29	30	24

Bar chart with the number of files per commit distribution

Changed files per commit





WordCloud of git authors

	word	frequency
0	Steven van Beelen	4357
1	Allard Buijze	3156
2	smcvb	822
3	Mateusz Nowak	506
4	Milan Savic	449
5	Mitchell Herrijgers	438
6	Gerard Klijs	247
7	Marc Gathier	231
8	sara pellegrini	163
9	Rene de Waele	129

Wordcloud of git authors

